Rajdeep Chakraborty

rajdeep.ch138@gmail.com
LinkedIn GitHub Website

Research Interests

Computational Psychiatry, Social Cognition, Decision-Making, Neurotechnology.

Education

Maulana Abul Kalam Azad University of Technology *2018–2022*
Bachelor of Technology in Mechanical Engineering Kolkata, India

Research Experience

European Molecular Biology Laboratory (EMBL) *Feb 2026–Present*
Scientific Trainee Rome, Italy

- Independently analysing a PhD project on motivational conflict and social defeat, including computational modelling of decision-making in uncertain environments using partially observable Markov decision processes (ongoing).
- Developed Python-based workflows to process and analyse multimodal datasets, including mouse tracking, fibre photometry, and calcium imaging, and produced publication-quality figures.
- Built a DeepLabCut pipeline using segmentation models to significantly reduce missing tracking data across 385 recordings and enabled spatial classification of mice in behavioural assay.
- Conducted mouse experiments and performed wet-lab work, including cryosectioning and tissue mounting.

Indian Institute of Science *Jul 2024–Apr 2025*
Research Assistant Bengaluru, India

- Designed and implemented a novel Arduino-based automated assay to study mouse decision-making under varied stress-reward contingencies, enabling experimenter-independent control of trials.
- Built a Python pipeline for automated data logging, backup, session-wise performance tracking, and behavioural visualisations.
- Performed behavioural experiments and conducted cryosectioning and histological processing of mouse brain tissue.

Skills

Programming & Data Analysis: Python, MATLAB, GraphPad Prism, Fiji/ImageJ
Modelling & Machine Learning: POMDP, Image Segmentation, Model evaluation
Pose Estimation & Behavioural Analysis: DeepLabCut, SLEAP, TRex, SimBA
Computing: HPC cluster computing (SLURM), Git, Shell scripting
Hardware & Embedded Systems: Arduino scripting, circuit electronics

Volunteering Experience

Online Course Helper *Apr 2026*
European Molecular Biology Laboratory

- Supported participants in a three-day EMBL-wide Python course by answering questions during online sessions.

Student Outreach Volunteer *Sep 2024–Dec 2024*
Indian Institute of Science

- Delivered introductory neuroscience talks to school and undergraduate audiences and co-organised a department-wide outreach event.

Professional Experience

Packt Publishing *Oct 2023–Jun 2024*
Technical Editor Remote

- Reviewed and validated code in technical manuscripts, coordinating with authors and reviewers to ensure accuracy and timely delivery.

Focus Edumatics *Aug 2022–May 2023*
Online Mathematics Tutor Remote

- Delivered individualised tutoring to students in grades 5–12; maintained a 4.9/5 average rating across sessions.

Certifications

Neuromatch Academy: Computational Neuroscience *Jul 2023*
HarvardX: Fundamentals of Neuroscience *May 2023*
HarvardX: Using Python for Research *Sep 2022*